

Team Involvement in Demonstration

This document describes each team member's role, contributions, and specific responsibilities during the OptiCrop project demonstration.

Team Member Profiles

Name	Role	Core Responsibilities
Member 1 (Team Lead)	Project Manager & Backend Developer	Project coordination, sprint planning, Flask app architecture, app.py, risk management
Member 2 (ML Lead)	ML Engineer	Dataset analysis, model training & evaluation, model.pkl creation, EDA notebooks, metrics report
Member 3 (Frontend Dev)	UI/UX Developer	HTML templates, Bootstrap styling, JavaScript validation, mobile responsiveness, UX testing
Member 4 (QA & Docs)	QA Engineer & Tech Writer	Unit & integration tests (pytest), this documentation set, README, performance testing

Demo Day Role Assignment

Segment	Primary Presenter	Support
Opening Problem Statement	Member 1 (Lead)	Member 2
Solution Architecture (slides)	Member 2 (ML)	Member 1
Live Demo 1 Rice Prediction	Member 3 (UI)	Member 1
Live Demo 2 Coffee Prediction	Member 1 (Lead)	Member 3
Feature Walkthrough	Member 3 (UI)	Member 4
Model Accuracy & Metrics	Member 2 (ML)	Member 4
Testing Evidence	Member 4 (QA)	Member 2
Scalability & Future Plan	Member 1 (Lead)	All
Q&A; Facilitation	Member 1 (Lead)	All

Individual Contribution Summary

Member 1 Team Lead & Backend Developer

- " Architected the Flask application structure (app.py, routes, error handlers)
- " Led all sprint planning sessions and maintained the project Kanban board
- " Wrote the Solution Architecture and Project Planning documentation
- " Managed GitHub repository branch strategy, code review approvals, releases

Member 2 ML Engineer

- " Downloaded, cleaned, and explored the Kaggle crop dataset (2,200 samples)
- " Trained and evaluated all 5 ML algorithms; selected Random Forest (99.09%)
- " Authored 01_Model_Training.ipynb with full reproducibility
- " Generated model.pkl and scaler.pkl; documented feature importance findings

Member 3 Frontend Developer

- " Designed and implemented index.html and result.html using Bootstrap 5
- " Wrote validate.js for client-side input validation with real-time feedback
- " Conducted usability testing with 5 participants; implemented feedback
- " Ensured mobile responsiveness at 375px, 768px, and 1280px viewports

Member 4 QA Engineer & Technical Writer

- " Wrote 17 pytest unit and integration tests; achieved 100% pass rate
- " Conducted performance benchmarking with JMeter (100 concurrent users)
- " Authored all 18 documents in this project submission package
- " Maintained requirements.txt, README.md, and installation guide

GitHub Contribution Statistics

Member	Commits	PRs Opened	PRs Reviewed
Member 1	42	8	6
Member 2	31	5	4
Member 3	38	7	5
Member 4	27	4	9 (most reviews)
Total	138	24	All merged to main